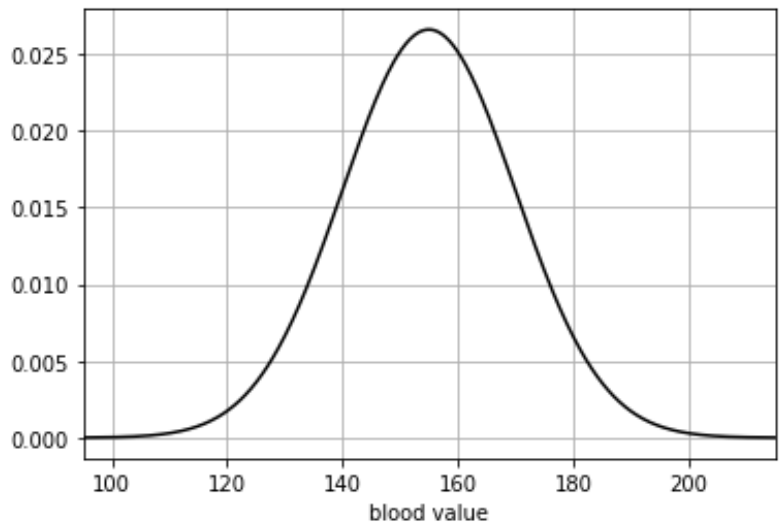
Lab: P-values

<u>standard frequentist way (not Bayesian):</u>

1) assume a **likelihood function *L* as your model**, aka **null hypothesis H_0**
2) take a datapoint x_0
3) calculate the ***probability P*** given ***L*** i. e. **given H_0** is true, **that x_0 has this value** or a **more extreme value**
4) accepting or rejecting H_0 based on ***P*** and the threshold α

example: 1) a healthy person has a blood value that follows a **normal distribution** with $\mu = 155, \sigma = 15$, i. e. $\boldsymbol{H_0}: N(\boldsymbol{\mu = 155}, \boldsymbol{\sigma = 15})$



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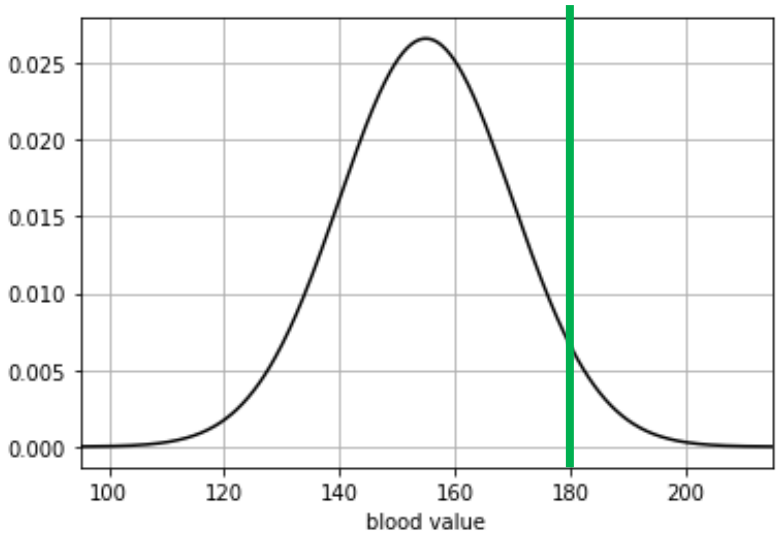
2) a patient has the value x_0 **= 180**



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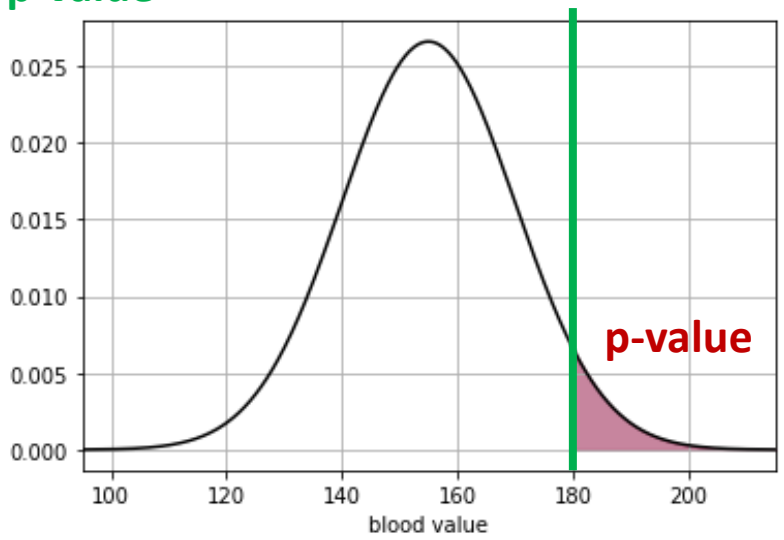
3) probability $\boldsymbol{P(x \geq 180 | H_0) = 0.048}$ called **p-value**



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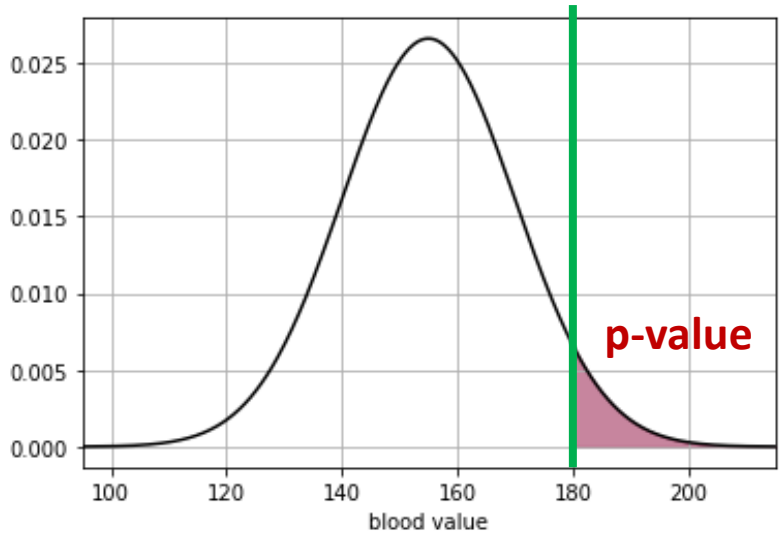
4) for **α = 0.05**, H_0 is **rejected**, i. e. patient is **not healthy**

→ alternative hypothesis $\boldsymbol{H_1}$: **not healthy**



Lab: P-values

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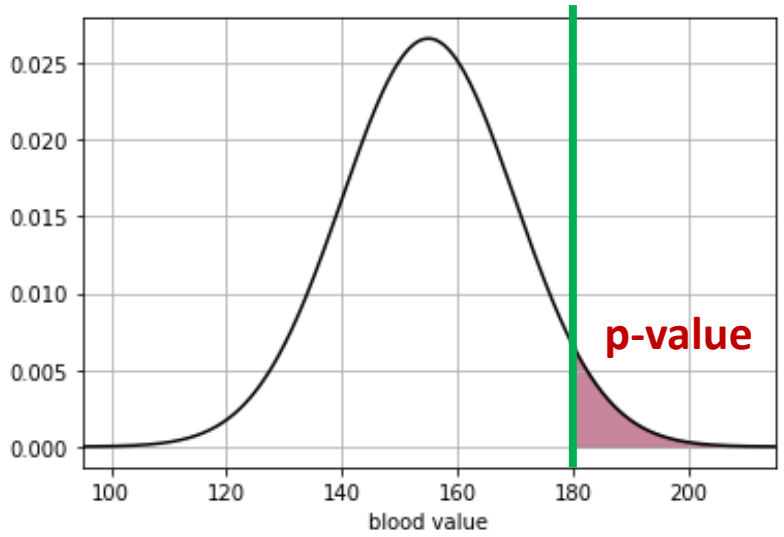
→ alternative hypothesis $\boldsymbol{H_1}$: **not healthy**

We just performed a so-called Z – Test
(comparing one value to a normal distribution)



Lab: P-values

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caveats:

- the model ***L*** for H_0 has to **be known**
- the p-value $P(x \geq x_0|H_0)$ **does not** tell if H_0 is true or not
- the p-value $P(x \geq x_0|H_0)$ **does not** tell which hypothesis is more likely
- the p-value just gives $P(x \geq x_0|H_0)$
- the threshold α for accepting/rejecting H_0 is **arbitrary**
- we are aiming on disproving a hypothesis, by assuming it is true, **without** leading to a contradiction

Lab: P-values

The Meaning of P

- four step exercises

The_Meaning_of_P.ipynb
The_Meaning_of_P_I.ipynb
The_Meaning_of_P_II.ipynb
The_Meaning_of_P_III.ipynb

- each step:
  - we discuss the problem set to be solved
  - you ask questions
  - you will **work in groups in breakout rooms**
  - after time is up: return to the **main room** and **discuss the solution**
  - you will **get the sample solution** and **compare it to your own solution**
  - material for **next step** will be pushed to **277A repo**
  - repeat: next step

- You don’t need to turn anything in. You will have the complete solution at the end.

Lab: P-values

The Meaning of P

0: read and try to understand the introduction part

1) The p-value

The standard workflow is the following:

- you run a statistical test
- this test returns a p-value
- if the p-value is below a certain value, say $\alpha = 0.01$, then the test is regarded significant

For example, we want to compare two samples A and B, which - in order to keep the example simple - we generate via random, normally distributed numbers with the same mean and the same variance:

```
mu   = 0
sig1 = 1
sig2 = 3
N1   = 50 # size sample A
N2   = 80 # size sample B
```

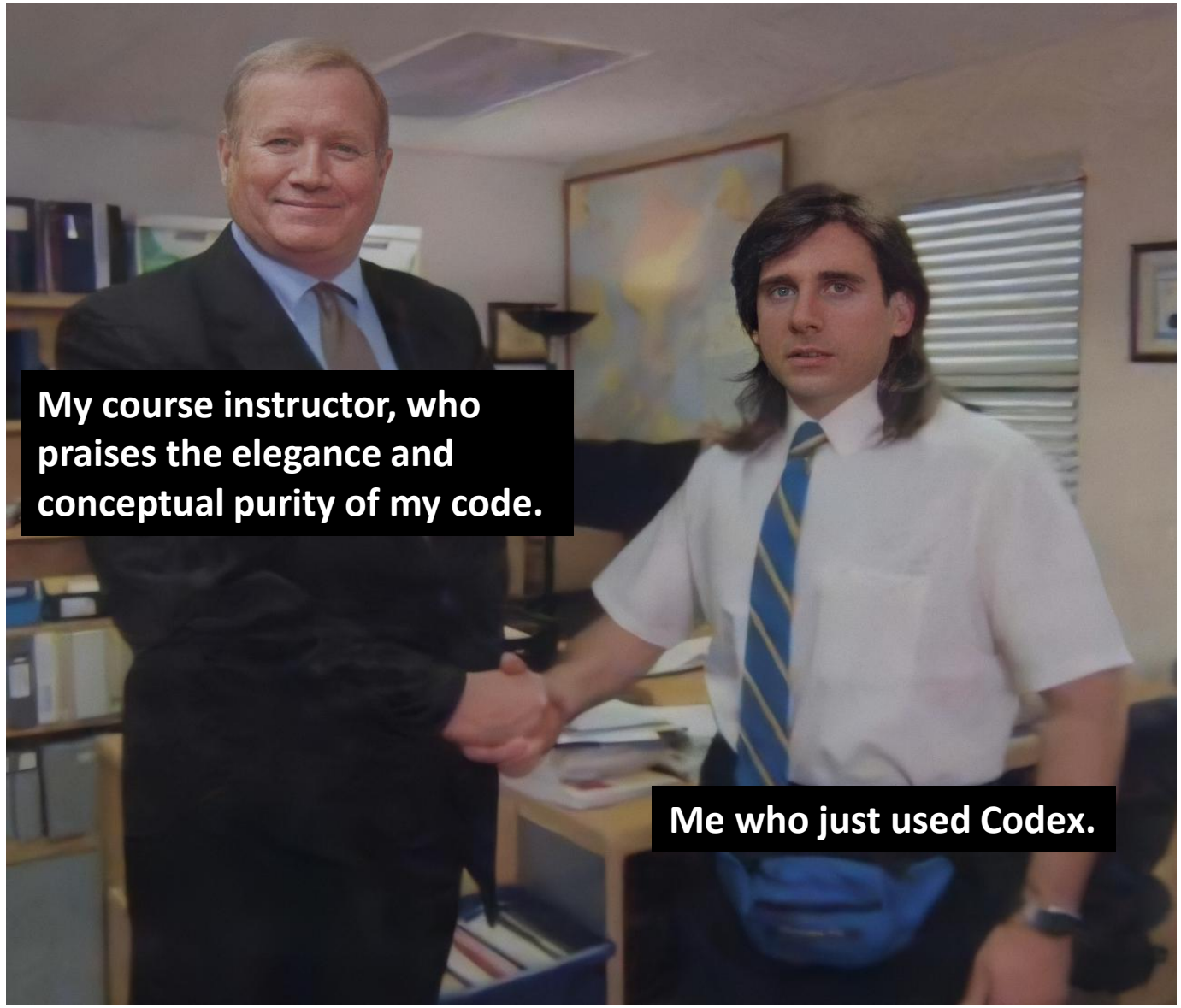
```
np.random.seed(43) # make results comparable
```

```
sampleA, sampleB = np.random.normal(mu, sig1, (N1,),), np.random.normal(mu, sig2, (N2,))
```

Lab: P-values

The Meaning of P

Enjoy :)



Lab: P-values

The Meaning of P

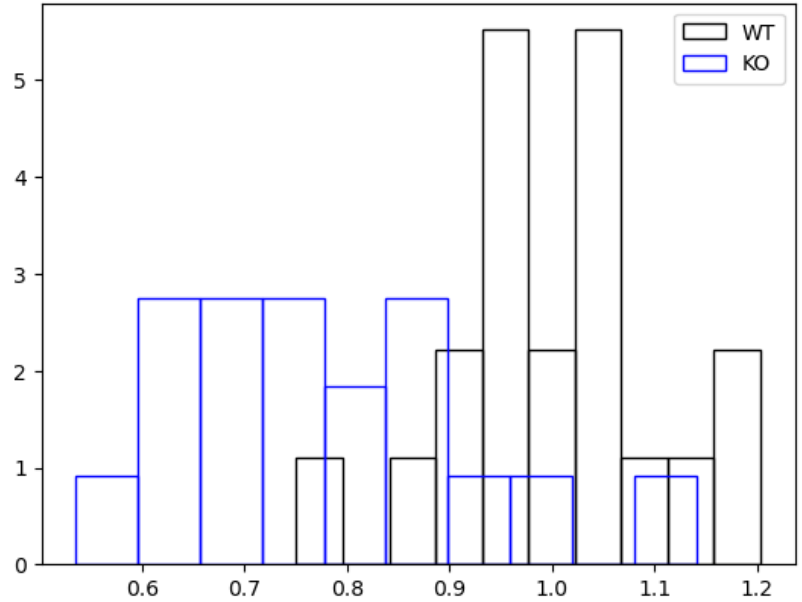
I:

2.1) Real Data

In an experiment, the normalized body weight of genetically modified mice ("KO") was compared to the body weight of healthy ("WT") mice.

```
WT = np.array([0.9110807, 1.088919, 1.03329, 0.9699879, 0.9593626, 1.03736, 0.9779246, 0.9400842, 1.030455, 1.051536, 1.183377, 0.9
KO = np.array([0.6693618, 0.7789116, 0.7524274, 0.6443067, 1.140946, 0.9826416, 0.6675799, 0.6960907, 0.8166829, 0.616216, 0.885979
```

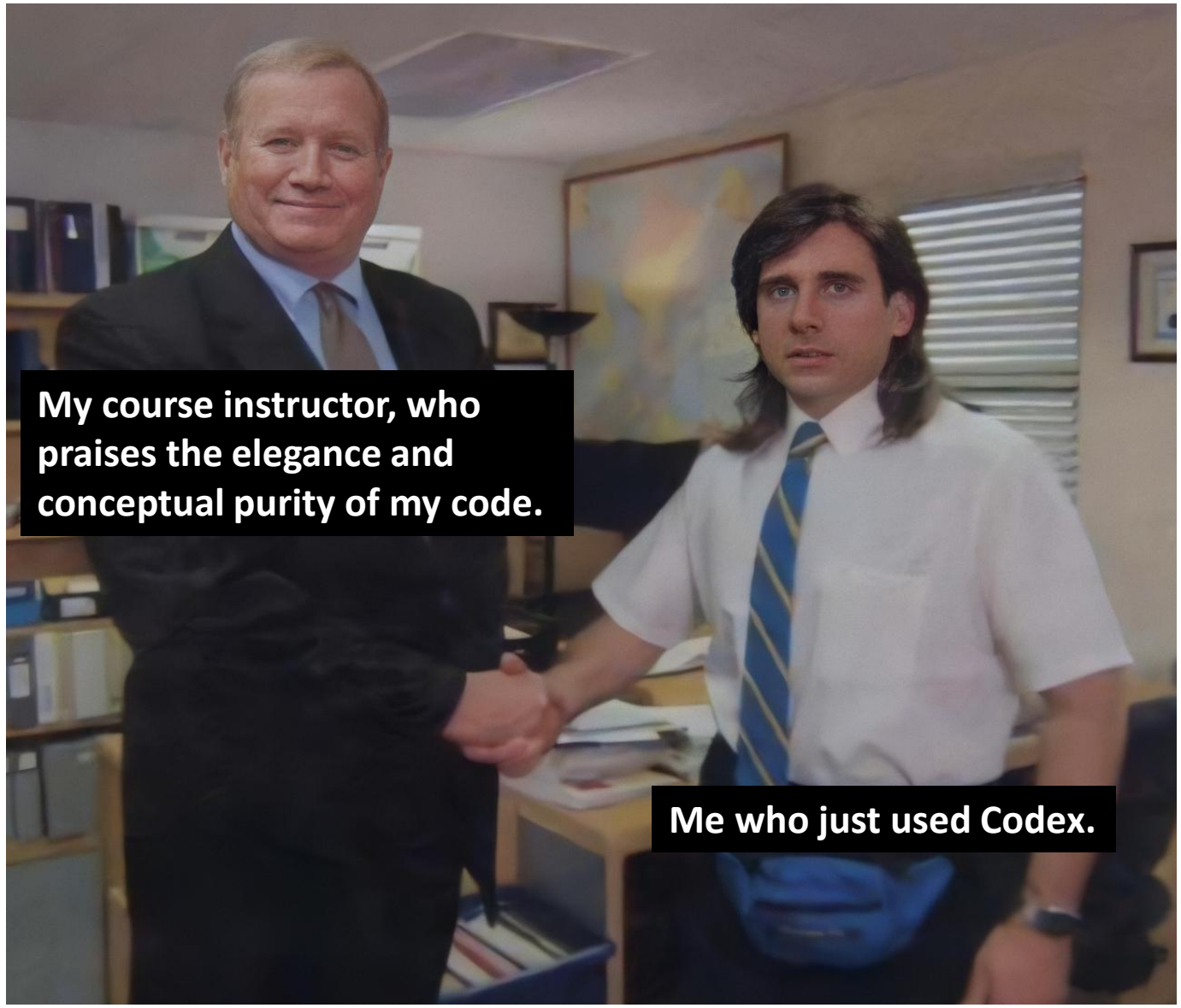
- write a small function **t_Test** using `def` that runs a t-test on the data and returns the p-value as output and as print statement. What is the result?
- plot a histogram of the two samples like the one above
- according to its definition: what is the largest and what is the smallest value of a p-value possible?



Lab: P-values

The Meaning of P

Enjoy :)



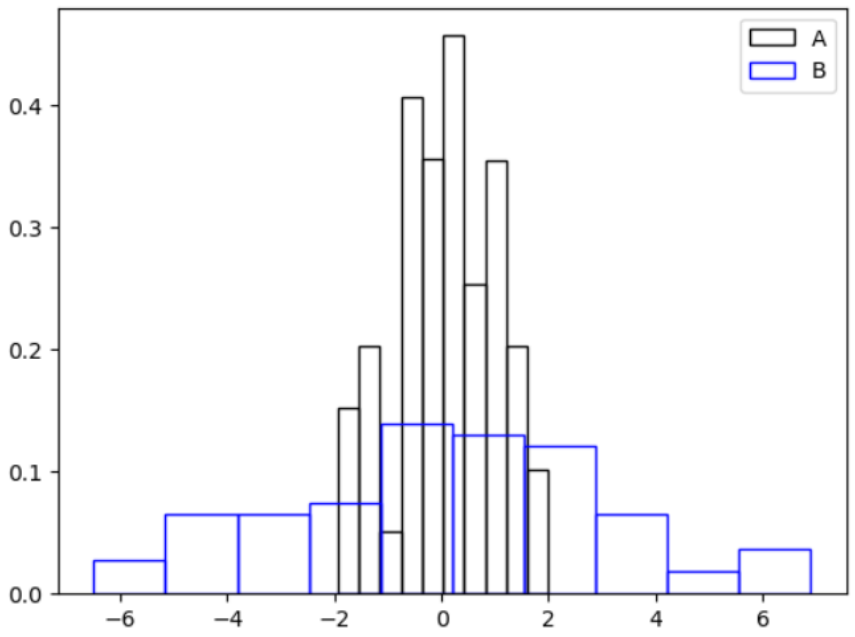
The Meaning of P

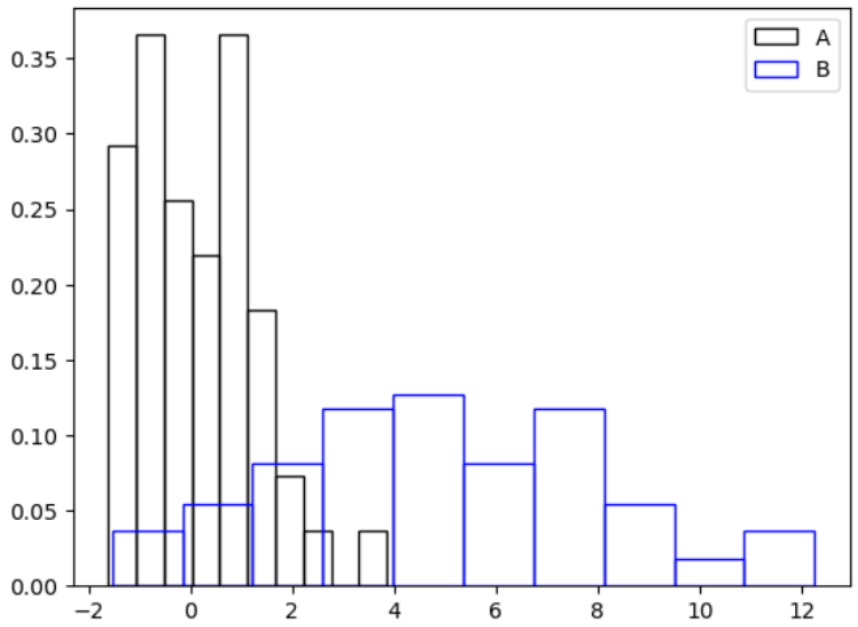
II:

2.2) How does the p-value change if we change μ and σ?

In order to understand how the p-value changes if the samples are different, generate two samples A and B like in Section 1, but now:

- change μ of one sample, say sample B, and run the t-test for a set of five different values for μ_B
- set $\mu_A = \mu_B$ and run the t-test for a set of five different variances for σ_B^2
- print the p-values each time. How do they change?

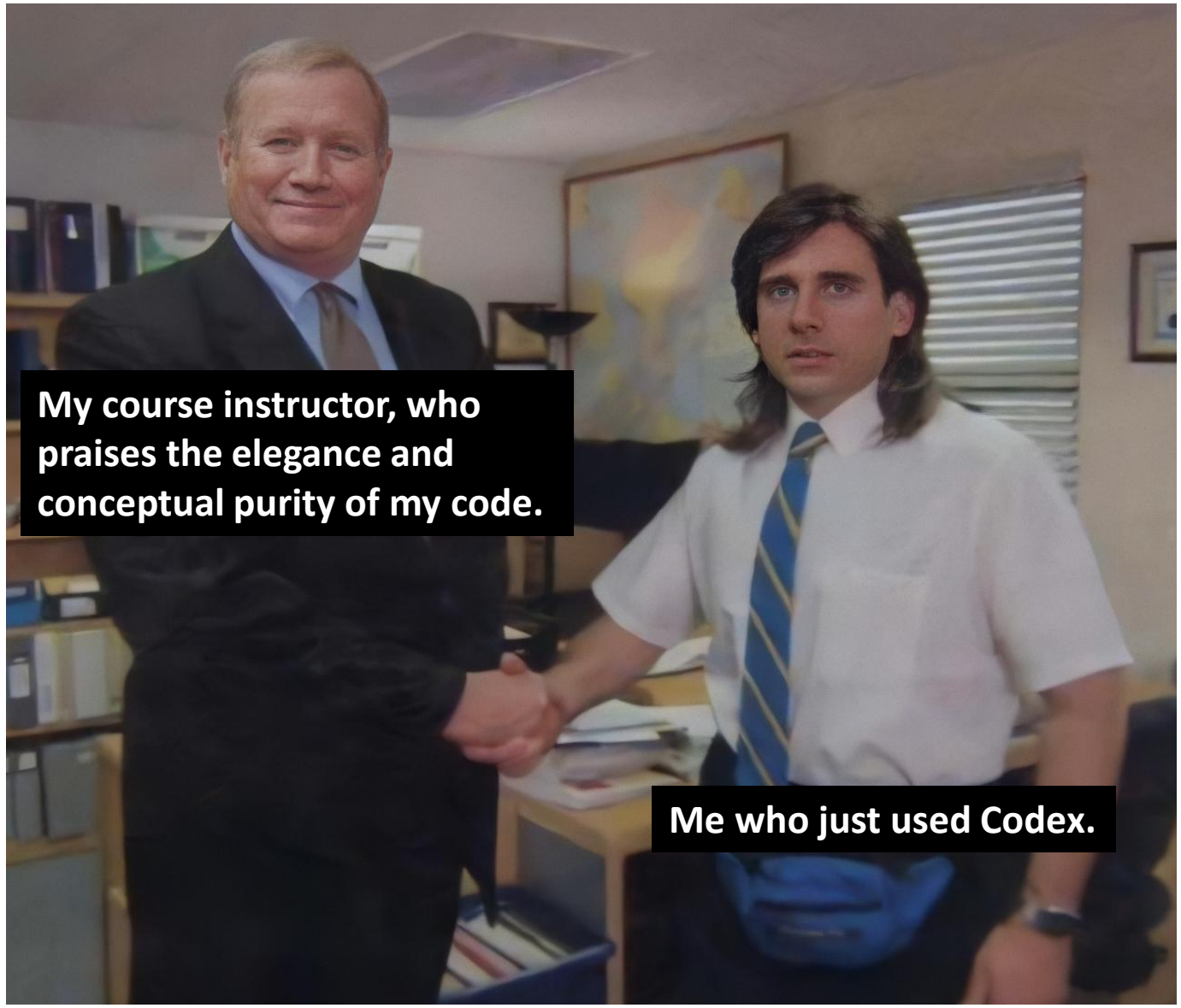




Lab: P-values

The Meaning of P

Enjoy :)



The Meaning of P

III: 2.3) Multiple Testing

Next, we want to simulate that we are running a t-test on each of $N = 1000$ genes of two samples A and B. There are two scenarios:

1. H_0 is true: Sample A and B are both from untreated mice. Each sample has three replicates (see diagram below). Each gene $i = 1, \ldots, N$ has a different expression level $L_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, but a given gene $i = k$ has the same expression level for sample A and B. Generate the dataset randomly by drawing the expression values from $\sim \mathcal{N}(\mu_i, \sigma_i^2)$ and run a t-test on each gene and plot the histogram of the 1000 p-values. What does the p-value distribution look like?
   Hint: for the simulation we have to choose μ and σ randomly.
   Start with eg. `Mus, Sigmas = np.random.uniform(0,1,(N,)), np.random.uniform(0.1,4,(N,))`

2) H_0 is false: Sample A came from untreated mice and sample B from treated mice. Therefore, in 20 cases, the expression level of a given gene $i = k$ is 2.5 times higher in A then in B. Generate the dataset and run a t-test and plot the histogram of the 1000 p-values. What does the p-value distribution look like now?

Lab: P-values

The Meaning of P

III:

scenario 1) H0 = true
sample A and B are both untreated

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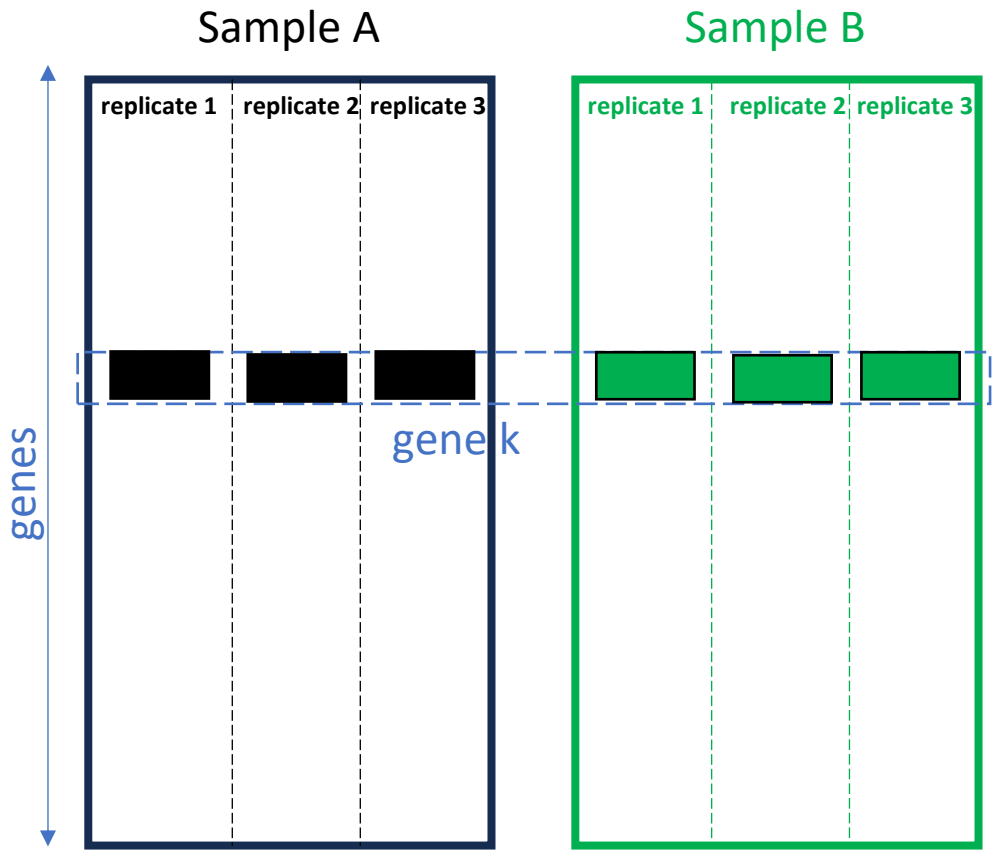
```
np.random.normal(m_k,
                 s_k,
                 (3,))
```




```
np.random.normal(m_k,
                 s_k,
                 (3,))
```

Lab: P-values

The Meaning of P

III:

scenario 2) H0 = false
sample A untreated and sample B treated

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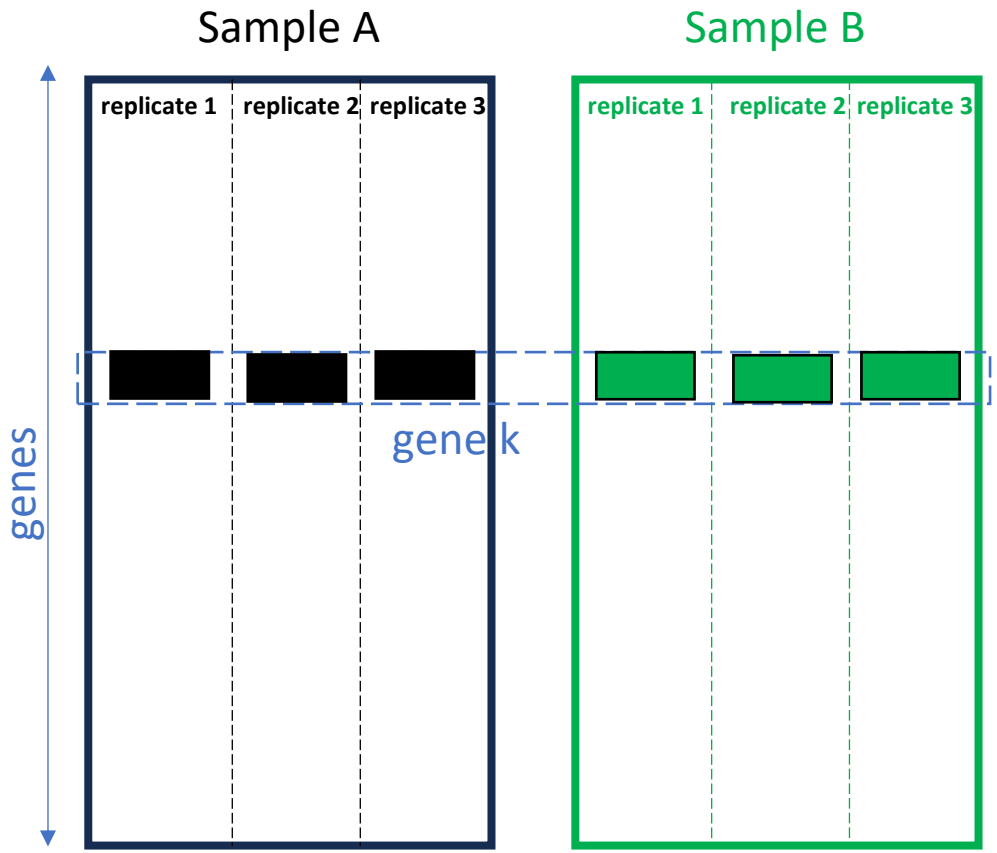
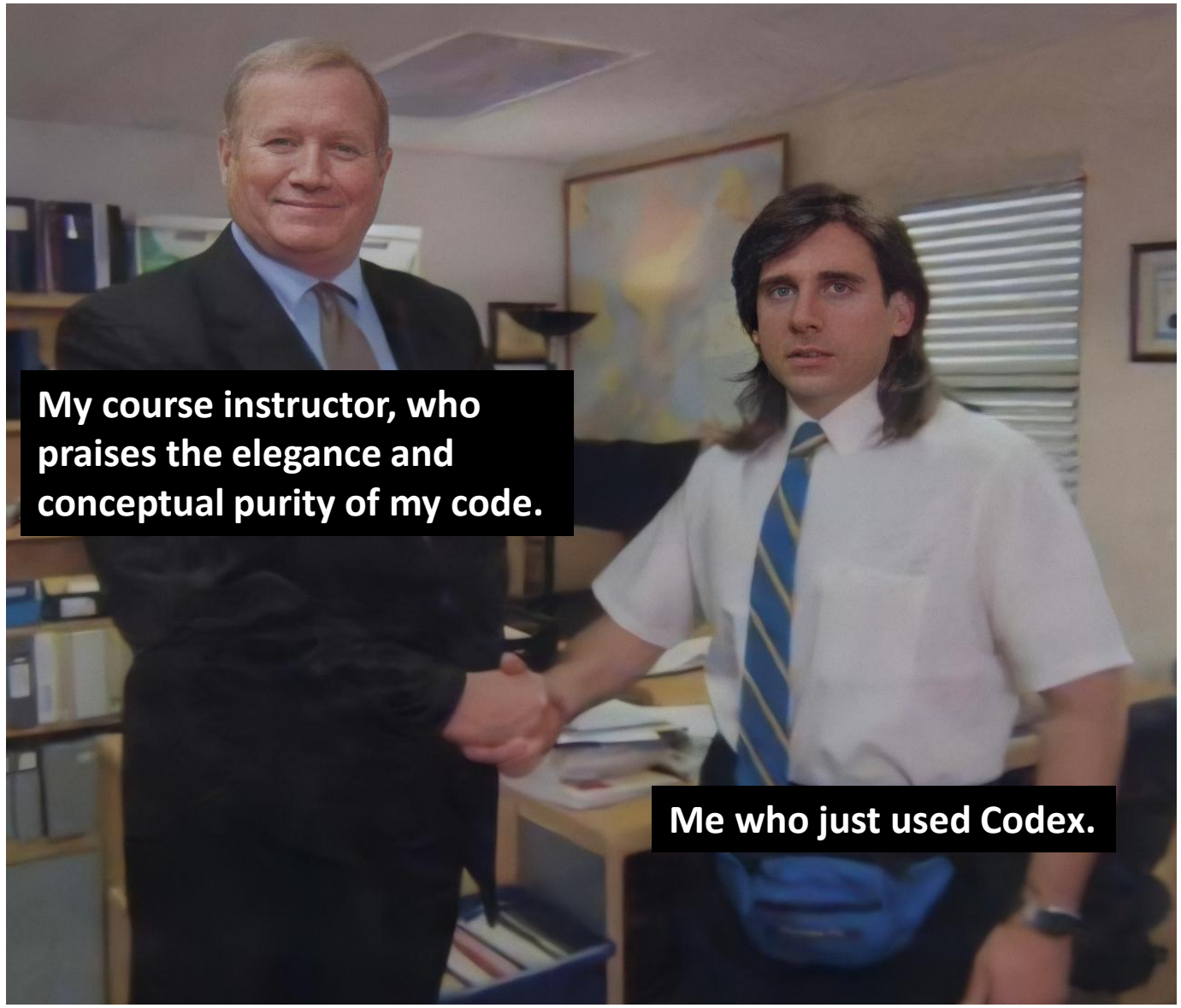
```
np.random.normal(m_kA,
                 s_kA,
                 (3,))
```




```
np.random.normal(m_kB,
                 s_kB,
                 (3,))
```

note: results might vary due to randomness
→ try also N = 10,000 or 100,000

Lab: P-values

The Meaning of P

Enjoy :)